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09/12/24 MIS

[22EE51]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER -2024

AC ROTATING MACHINERY

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	A 3-phase, 10 HP squirrel cage induction motor is wound for 6 poles. When the motor is connected to 230 V, 50 Hz supply, at full-load, it operates at 4% slip. Determine (i) full load speed. (ii) full load torque in Newton-metre.	L3	CO1	**02**
	b.	Why synchronous motor always runs at synchronous speed ?	L3	CO2	**02**
	c.	Draw the equivalent circuit of a three phase synchronous generator.	L2	CO3	**02**
	d.	List the different methods used to determine voltage regulation of an alternator.	L1	CO5	**02**
	e.	State the necessary conditions for parallel operation of alternators.	L1	CO6	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Draw the complete torque-slip characteristics of an induction motor and show starting torque and breakdown torque. Mark the stable and unstable regions.	L2	CO1	**05**
		OR			
	b.	A 3φ, 15 HP, 460 V, four-pole, 60 Hz, 1728 rpm induction motor delivers full output power to a load connected to its shaft. The windage and friction loss of the motor is 750 W. Determine the **i.** Mechanical power developed. **ii.** Air gap power. **iii.** Rotor copper loss.	L3	CO1	**05**
3.	**a.**	How will you make a single-phase induction motor self-starting?	L2	CO2	**05**
		OR			
	b.	Describe the construction and working of a single-phase shaded pole induction motor.	L2	CO2	**05**

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		PART-C			6 X 10=60
4.	a.	A three-phase, 460V, 1740rpm, 60Hz, four-pole wound-rotor induction motor has the following parameters per phase: $R_1=0.25\Omega$, $X_1 = X'_2=0.5\Omega$, $R'_2 = 0.2\ \Omega$, $X_m=30\Omega$. The rotational losses are 1700 watts. With the rotor terminals short-circuited, find **i.** Starting current when started direct on full voltage. **ii.** Starting torque.	L3	CO3	**05**
		OR			
	b.	Explain how the equivalent parameters of three phase induction machine be determined by no load and blocked rotor tests.	L2	CO3	**05**
5.	a.	Define the three modes of operation of an induction machine: motor mode, generator mode, and braking mode. What are the key characteristics that distinguish these modes?	L2	CO4	**05**
		OR			
	b.	Compare an induction generator with an alternator.	L3	CO4	**05**
6.	a.	Derive the EMF equation of alternator.	L2	CO5	**05**
		OR			
	b.	From the following test results, determine the voltage regulation of 2000V, single phase alternator delivering a current of 100A at **i.** unity p.f **ii.** 0.8 leading p.f Test results: Full load current of 100A is produced on short circuit by a field excitation of 2.5 A. An emf of 500V is produced on open-circuit by the same excitation. The armature resistance is 0.8 Ω.	L3	CO5	**05**
7.	a.	Describe the sub-transient, transient & steady state reactance of a synchronous machine. Obtain an expression for the synchronizing power of a synchronous machine.	L3	CO6	**05**
		OR			
	b.	Explain the different methods of synchronization of an alternator with bus bar.		CO6	**05**

8.	a.	How a rotating magnetic field is generated in a 3- Ø Induction Motor. Explain in detail with necessary diagrams and equations.	L2	CO1	**10**
		OR			
	b.	Discuss the phenomena of Cogging or magnetic locking and Crawling in an induction motor.	L2	CO1	**10**
9.	a.	Describe any two speed control methods for three-phase induction motor, with appropriate diagrams.	L2	CO2	**10**
		OR			
	b.	Why a starter is required to start a 3-phase induction motor? Explain the working of a direct-on-line starter with neat sketch.	L2	CO2	**10**
10.	a.	Explain the experimental method to determine the equivalent circuit parameters of a single phase induction motor.	L2	CO3	**10**
		OR			
	b.	Explain in detail equivalent circuit and phasor diagrams of a synchronous generator for lagging power factor, unity power factor and leading power factor.	L2	CO3	**10**
11.	a.	Sketch the circle diagram and explain how it is useful in predetermining the performance of an Induction Machine.	L2	CO4	**10**
		OR			
	b.	No-load and blocked rotor test were performed on a 400 V, 3-phase delta-connected induction motor and the following results were obtained: No-load test: 400 V; 2.5A; 600 W Blocked-rotor test: 200 V; 12.5 A; 1500 W Determine energy (or working) component and magnetising component of no-load current, no load power factor, exciting resistance and reactance per phase referred to stator (primary) side assuming that friction and winding losses are 180 W. Also find short circuit current and power factor.	L3	CO4	**10**
12.	a.	What is V and Inverted-V curves of a synchronous motor? Explain in detail. Draw and explain the experimental set-up to obtain the	L2	CO5	**10**

		curves.			
		OR			
	b.	A 3-phase, 4-pole, star connected alternator has a smooth cylindrical type rotor. The effective resistance and synchronous reactance per phase are 0.15Ω and 2.5Ω respectively. Calculate the voltage regulation when delivering 250 A at 6.6 kV at different power factors of (i) 0.6 pf lagging. (ii) UPF (iii)0.8 pf leading.	L3	CO5	**10**
13.	**a.**	Explain any one of indirect methods of finding the regulation of a synchronous generator from the open circuit and short circuit characteristics.	L2	CO5	**10**
		OR			
	b.	Describe the parallel operation of three phase alternators with help of a neat diagram.	L2	CO6	**10**